EX NO:1	Butterworth IIR Filters
DATE:	1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

Program (Matlab Code):

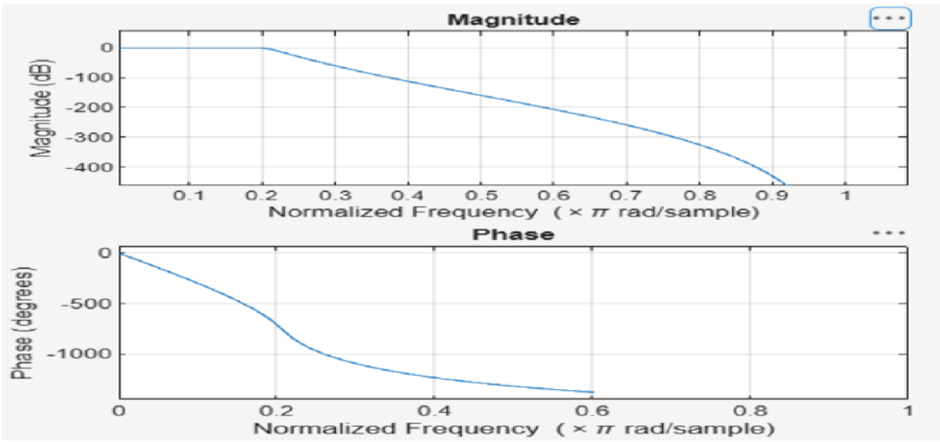
```
clc; clear
Fs=10000; Fp=1000; Fst=1500;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn);
freqz(b,a)
figure% Magnitude & Phase
[h,n]=impz(b,a,50);
...title('Impulse Response')
figure
stepz(b,a)
figure
zplane(b,a)
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

Objective 1:

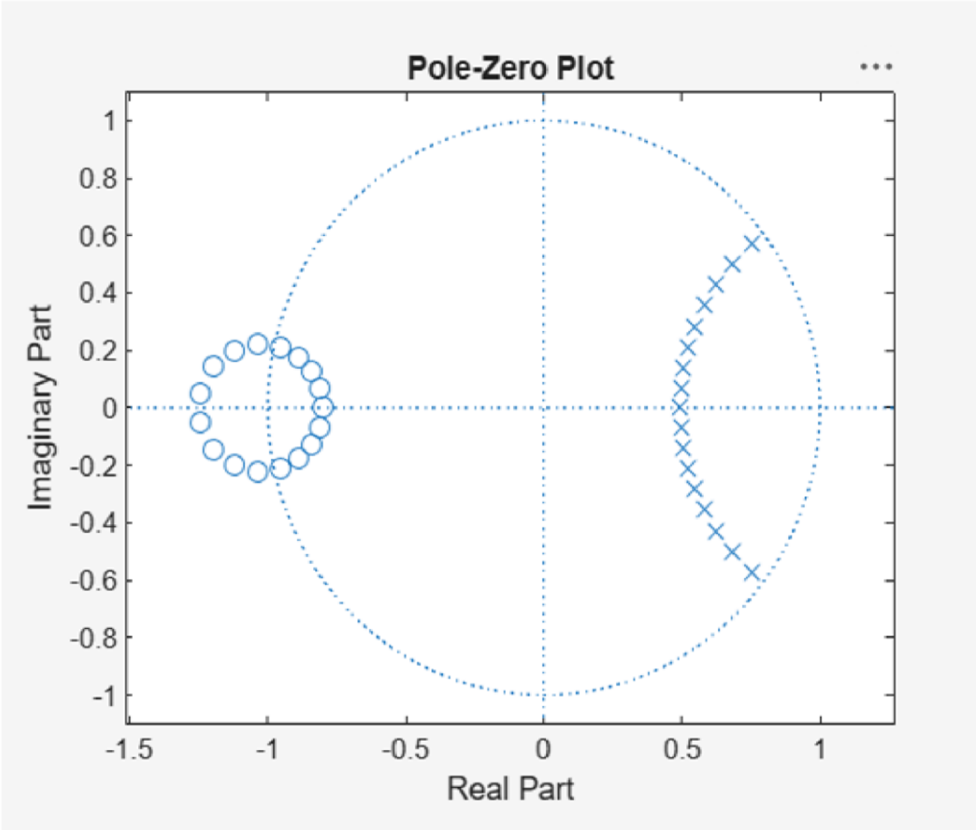
Minimum Filter Order (N) = 17

Cutoff Frequency (Wn) = 0.2083

Objective 2:

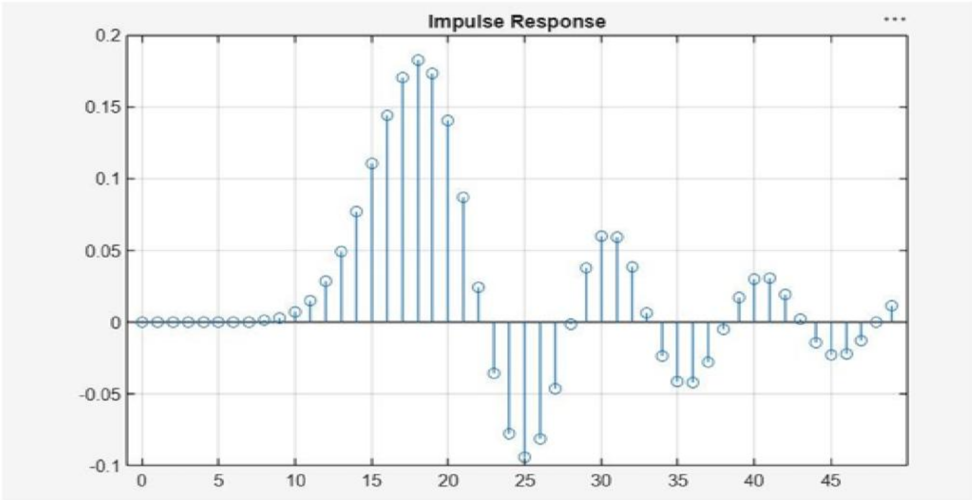


Objective 4 :

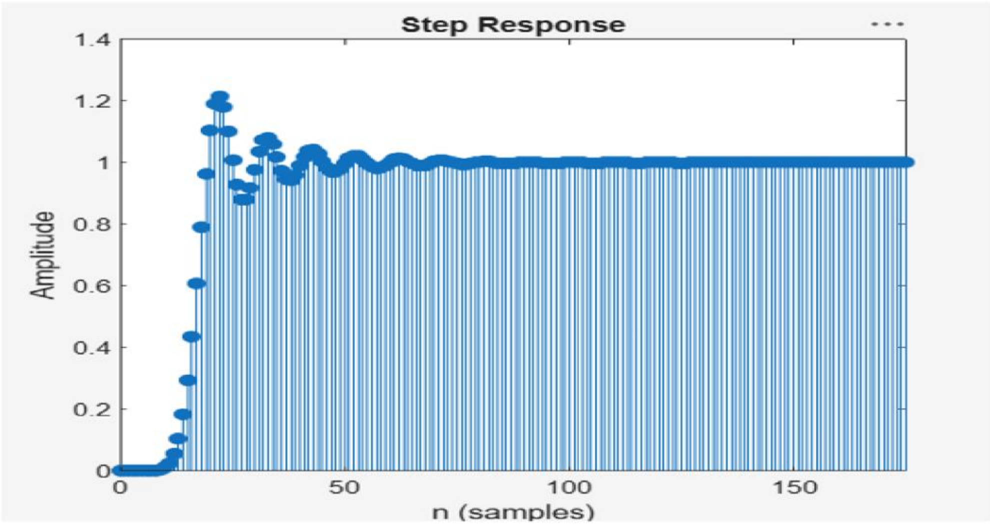


Pole-Zero Plot
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.5
-1
-0.5
0
0.5
1

Objective 3 :



Impulse Response
0.2
0.15
0.1
0.05
0
-0.05
-0.1
0
5
10
15
20
25
30
35
40
45



Step Response
Amplitude
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
50
100
150
n (samples)

EX NO:1	Butterworth IIR Filters
DATE:	1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

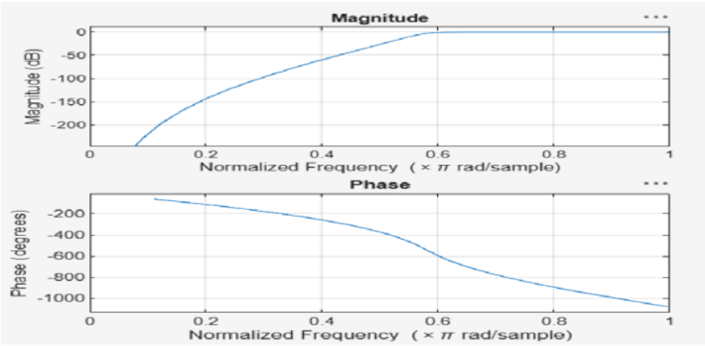
Program (Matlab Code):

```
Clc;
clear;
close all;
Fs=10000; Fp=3000; Fst=2000;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn,'high');
% Magnitude & Phase
freqz(b,a)
% Impulse Response
figure
[h,n]=impz(b,a,50);
stem(n,h), grid on
title('Impulse Response')
% Step Response
figure
stepz(b,a)
title('Step Response')
% Group Delay
figure
grpdelay(b,a)
title('Group Delay')
% Pole-Zero Plot
figure
zplane(b,a)
title('Pole-Zero Plot')
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

Objective 1 :

Minimum Filter Order (N) = 12
Cutoff Frequency (Wn) = 0.5807

Objective 2 :



Objective 3 :